

Extraction of Perfectly Passive and Causal Circuit Models from Noisy Data

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Abstract— A standard least-squares algorithm can be used to find the best rational polynomial fit for noisy complex data, however, the fitted polynomial coefficients are also complex. A causal frequency response requires all coefficients of powers of $s = j\omega$ to be real. This paper introduces, for the first time, a least-squares fit that satisfies this restriction. The elements of the corresponding lumped circuit are obtained from the fitted coefficients by closed-form solution. Solutions are presently available for a library of over 600 circuit models. Of those lumped circuit models that realize a good fit, perfectly passive models are identified and may be selected. Complete details of the specialized least-squares fit and several examples are provided. Much of the process has been automated in MATLAB, including extensive use of MATLAB's symbolic equation facility.

Keywords—Cauchy method, compact models, curve fitting, least-squares fit, model extraction, optimization, Pade, parameter estimation, rational polynomial, vector fitting.

I. INTRODUCTION

The last century has seen extensive use of lumped circuit models to represent microwave circuits. In the simplest approach, the model topology is based on expert knowledge of the structure being considered and is informed by measured data or EM (electromagnetic) results. Then various structure- and model-specific strategies provide initial guesses for the model parameters. Finally, circuit-simulation based optimization refines those parameters. This can be time consuming and end in failure.

When an optimal model topology is not known *a priori*, vector fitting [1] can be invoked. This technique uses a least-squares algorithm to find a set of poles and zeros that can be used to directly determine an equivalent circuit and the desired rational polynomial model.

A related approach, the Cauchy method, has seen significant work in the microwave field. Two recent applications involve interpolation and extrapolation of a limited data set [2] and [3]. The method may also determine the coupling matrix for a filter based on measured data, [4]. Results resemble vector fitting except that singular value decomposition (SVD) is performed on the measured data providing an independent estimate of the minimum number of poles and zeros required for an accurate model.

All the above techniques allow rational polynomials with complex coefficients. In contrast, the impedance or admittance of a circuit composed of real (i.e., not complex) valued lumped elements must have only real coefficients in the s -domain rational polynomial transfer function. Complex coefficients correspond to complex-valued elements as well as a non-causal impulse response because the real part of the transfer function is no longer an even function of frequency,

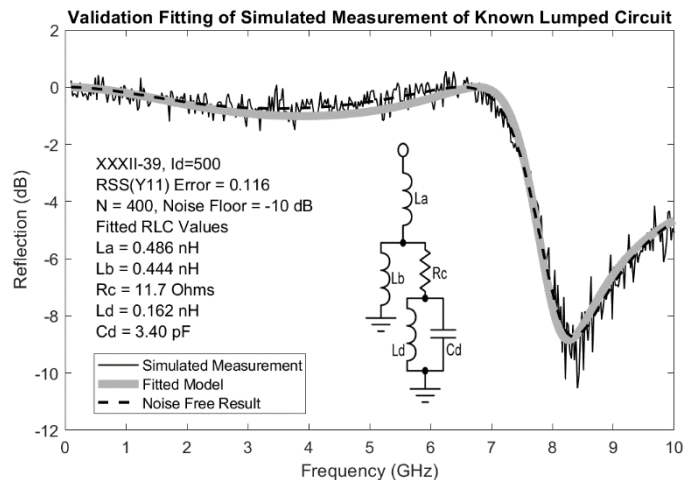


Fig. 1. For validation, we start with the indicated circuit, add noise to simulate measurement, and then select the best fitting model. The values of the original circuit are 0.5 nH, 0.4 nH, 10 Ohms, 0.2 nH, 3 pF. Among all 623 circuits in the base library, the original circuit provides the best fit.

and the imaginary part is no longer odd.

A different approach, [5], derives the rational polynomial transfer functions for over 600 circuit topologies. Then, for each circuit of N lumped elements, N simultaneous non-linear equations are formed by setting the transfer function equal to the data at N frequencies and solving for the N desired lumped elements. The model that best matches the entire simulation is then selected. Values of N up to eight have been used. This technique fails if there is noise in the data being fitted.

A similar technique has been used to develop perfectly causal distributed models for dispersive multiple-coupled transmission lines based on EM data [6].

In this paper, I describe, for the first time, a least-squares fit that extracts a perfectly causal rational polynomial best fit from noisy measured data. SVD could be used to discern the required polynomial order. However, because the least-squares fit is so fast, we perform the fitting for a variety of polynomial orders until we realize an acceptable result. This transfer function may be used directly to model the target data.

In some cases, a lumped model is preferred. In this work, transfer functions for 623 lumped circuits representing all possible circuits of up to five lumped elements have been symbolically derived and placed in the form of a rational polynomial. The least-squares fit described in the previous paragraph provides numerical values for these coefficients. The resulting simultaneous non-linear equations are then solved for the values of the lumped elements. Fig. 1 shows an example result that is discussed later in this paper.

This paper details the specialized least-squares algorithm, provides a validation example, and then applies the algorithm to a simulated measurement of a two-port chip resistor.

II. CAUSAL LEAST-SQUARES FIT

Our task is to find a rational polynomial of the form

$$z(\omega) = r(\omega) + jx(\omega) = \frac{a_M(j\omega)^M + \dots + a_1(j\omega) + a_0}{b_N(j\omega)^N + \dots + b_1(j\omega) + b_0} \quad (1)$$

that corresponds to a perfectly causal transfer function and yields a least-squares fit to a complex impedance data set, $z_i = r_i + jx_i$, measured at N_S radian frequencies, ω_i . Causality is guaranteed by forcing all a and b polynomial coefficients to be real. Simplifying notation, all matrix elements in (4) – (6) are summed over all measured data samples. The first term in (4) sums to N_S . The least-squares fit is realized by solving

$$\begin{bmatrix} A_A & A_B \\ A_B^T & B_B \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = [0] \quad (2)$$

where the superscript T indicates transpose and

$$a^T = [a_0 \quad a_1 \quad \dots \quad a_M], \quad b^T = [b_0 \quad b_1 \quad \dots \quad b_N] \quad (3)$$

$$A_A = \begin{bmatrix} 1 & 0 & -\omega^2 & 0 & \omega^4 & 0 \\ 0 & \omega^2 & 0 & -\omega^4 & 0 & \omega^6 \\ -\omega^2 & 0 & \omega^4 & 0 & -\omega^6 & 0 \\ 0 & -\omega^4 & 0 & \omega^6 & 0 & -\omega^8 \\ \omega^4 & 0 & -\omega^6 & 0 & \omega^8 & 0 \\ 0 & \omega^6 & 0 & -\omega^8 & 0 & \omega^{10} \\ \vdots & & & & & \ddots \end{bmatrix} \quad (4)$$

$$A_B = \begin{bmatrix} -r & x\omega & r\omega^2 & -x\omega^3 & -r\omega^4 & x\omega^5 \\ -x\omega & -r\omega^2 & x\omega^3 & r\omega^4 & -x\omega^5 & -r\omega^6 \\ r\omega^2 & -x\omega^3 & -r\omega^4 & x\omega^5 & r\omega^6 & -x\omega^7 \\ x\omega^3 & r\omega^4 & -x\omega^5 & -r\omega^6 & x\omega^7 & r\omega^8 \\ -r\omega^4 & x\omega^5 & r\omega^6 & -x\omega^7 & -r\omega^8 & x\omega^9 \\ -x\omega^5 & -r\omega^6 & x\omega^7 & r\omega^8 & -x\omega^9 & -r\omega^{10} \\ \vdots & & & & & \ddots \end{bmatrix} \quad (5)$$

$$B_B = |z|^2 A_A. \quad (6)$$

Dimensions of the matrices in (4) and (5) may be inferred from the size of their mating a and b vectors (3). Note that the full matrix (2) size is $M + N + 2$ by $M + N + 2$, i.e., the number of unknown coefficients needed for the rational polynomial (1). When there are known to be zero-valued coefficients in the transfer function, remove the rows and columns corresponding to the zero-valued coefficients.

To improve the signal-to-noise ratio, we work with spectrally dense data sets, sometimes even thousands of measured points. This is practical because this approach invokes much smaller matrices than SVD. SVD typically uses matrices with one dimension equal to the number of samples. Here, each matrix element is a sum over all samples. In addition, there are many duplicate and near-duplicate elements in (4) – (6), further reducing required memory and effort. Evaluating a small set of ‘sufficient statistics’ is all that is required to fill the matrix. These sufficient statistics need be evaluated only once when the same data is to be fitted to multiple transfer functions.

Note that all expressions in (2) – (6) evaluate to pure real values, thus insuring causality of the result, (1).

There are two special considerations for this fit. First, (2) is singular, there are an infinite number of solutions. Since (1) is a ratio, if we have one solution for the coefficients, a second solution is obtained by multiplying the first solution by an arbitrary constant. Thus, one of the coefficients in (3) must be set to a fixed value. Usually, we set $b_0 = 1$. However, if it turns out that the optimal value for b_0 is zero, some other coefficient must be selected. Once a coefficient is selected, substitute into (2) and manually solve for it. This reduces the dimension of the left-hand side of (2) by one and gives us a non-zero right hand side. The remaining coefficients are solved by Gaussian elimination or equivalent.

The second consideration is that the sufficient statistics that fill the matrices of (4) and (5) require sums of high powers of the radian frequency. With normal double precision, more than a small handful of coefficients can result in catastrophic numerical precision errors. Normalizing the data and frequencies going realizes a limited solution. However, I have found that true freedom from the numerical precision curse is realized only by increasing the precision of the arithmetic used in the algorithm. Fortunately, MATLAB provides just such a capability. I invoke extremely high precision for my implementation of this algorithm and all precision problems evaporate. Due to the small size of the required matrices (for example, 101 coefficients require only a single 100×100 matrix solve), no noticeable impact on fitting speed has been seen. Increased precision is also advantageous when evaluating the fitted transfer function (1).

III. COMPONENT LIBRARY

I have set up and solved for all possible one-, two-, three-, four-, and five-element lumped components (i.e., circuits composed of resistors, capacitors, and inductors connecting two external nodes). I have excluded all obvious degenerate circuits, e.g., two resistors in parallel do not have a unique solution matching an equal valued single resistor. However, there are non-obvious degenerate circuits that I include as separate components. When these are encountered, a model may be simplified by substituting the simpler component.

Several components are equivalent, i.e., each component can be transformed into the other with identical transfer functions. When possible, equivalence equations have been solved using MATLAB’s symbolic equation facility. There may be equivalences between some components for which MATLAB’s ‘solve’ function fails to find a solution.

All components are solved for impedance, admittance, poles, zeros, any equivalencies, and synthesis equations. Synthesis equations take as input the rational polynomial coefficients that result from fitting the component to the measured data. Their result is the R, L, and C values that correspond to those polynomial coefficients.

Synthesis equations for several five-element components appear to have no solution. Each of these component topologies resembles a Wheatstone Bridge. It has been proven that a general solution to the quintic equation in terms of addition, subtraction, multiplication, division, and square root

Table 1. A transfer function signature indicates which coefficients of a solved transfer function are non-zero. There are presently 53 unique signatures in the base library of 623 lumped circuits.

Library of Transfer Function Signatures			
1-1	1010-11111	111-101	1111-1111
1-10	10101-1010	111-11	1111-11110
1-11	10101-101010	111-110	1111-11111
10-1	10101-11111	111-111	11110-1111
10-101	101010-10101	111-1110	11110-11111
10-11	11-1	111-1111	11111-1010
10-111	11-10	1110-111	11111-10101
101-10	11-11	1110-1111	11111-1110
101-1010	11-110	1110-11111	11111-1111
101-111	11-111	1111-101	11111-11110
101-1111	110-11	1111-1010	11111-11111
1010-101	110-111	1111-110	
1010-10101	110-1111	1111-111	
1010-1111	111-10	1111-1110	

does not exist [7]. This might be the reason no solution was found for these components.

In [5], all solutions are derived manually for up to four elements. Here, all solutions are obtained using the symbolic math functionality in MATLAB. A nodal net list is specified for each component, and then everything else is set up and solved automatically and symbolically in a MATLAB script.

Among the 623 components, there are 53 unique transfer function signatures, Table 1. For example, the signature ‘1010-101’ corresponds to the transfer function

$$y(\omega) = \frac{a_3(j\omega)^3 + a_1(j\omega)}{b_2(j\omega)^2 + b_0}. \quad (7)$$

There are nine components in the library whose transfer functions have this signature. Two of them are equivalent: a capacitor in parallel with a series LC, and a capacitor in series with a parallel LC. Equations for converting between these two components have been solved. The other seven components have either four or five elements and are degenerate in that they simplify to either of these three-element components.

IV. MODEL EXTRACTION

Model extraction is a three-step process. First, obtain spectrally rich measured data for an impedance or admittance. Result accuracy increases substantially while least-squares fitting time remains short even for thousands of data points. Second, perform the polynomial fitting described in the previous section. If the resulting polynomial lacks sufficient accuracy, increase the number of terms. Find the required number of terms by trial and error, or if desired, by using SVD.

In some cases, the fitted polynomial is sufficient. If a lumped model is desired, fit the required polynomial (Table 1) for each lumped circuit in the library. Then use the stored synthesis equations to determine the corresponding RLC values. Evaluate each of the components in the library and select the best model/fit.

For passivity, we require the real part of the impedance or admittance transfer function to be non-negative at all frequencies. To determine passivity, evaluate the symbolic

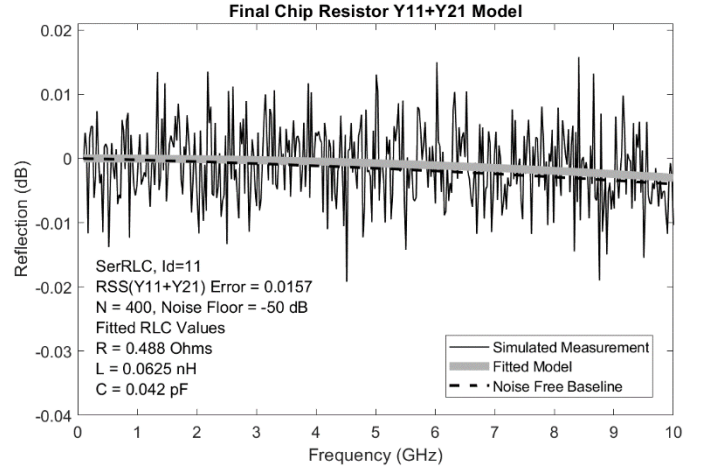


Fig. 2. The admittance of the port 1 to ground portion of the π -network model, y_{11} , is equal to $Y_{11} + Y_{21}$. Noise is added to the EM analysis of the chip resistor to simulate a measurement. This is the best resulting passive model.

transfer function for $s = j\omega$, take the real part of the numerator multiplied by the complex conjugate of the denominator, and solve for all zeros. If any zero is pure real, (zero frequency excluded), then the transfer function is not passive. This is easily realized symbolically in MATLAB.

V. VALIDATION

The extraction is tested by taking a known circuit, adding noise to simulate a measurement, extracting the underlying lumped model, and comparing to the original circuit. Fig. 1 shows the result for arbitrarily selected circuit number 500 taken from the library of 623 circuits. Noise was added to bring the ‘noise floor’ of the data to -10 dB and models were extracted for all 623 circuits. Circuit 500 provides the best fit.

VI. TWO-PORT CHIP RESISTOR MODEL EXTRACTION

A typical extraction for multi-port circuits converts S-parameter measurements to Y-parameters. The Y-parameters determine a π -network model [5]. Model extraction is performed for each π -network admittance with a two-port lumped model assembled from the results. An analogous approach uses Z-parameters and a T-network model [8].

Illustrating with a chip resistor from Sonnet example files [9], [10], we add noise to an EM result, simulating a -50 dB noise floor. Fig. 2 shows the best fit passive model for the input-to-ground leg, $y_{11} = Y_{11} + Y_{21}$, of the π -network. Surprisingly, it is a series RLC. Because more complicated circuits all result in less favorable fits, we infer that searching for a more complicated model is pointless. Fig. 2 shows the original EM data for the chip resistor (‘Noise Free Baseline’), the simulated measured data, and the fitted model. The fitting and RSS error is evaluated in terms of admittance. The plotted reflection coefficient corresponds to the admittance.

If perfect passivity is not required, the y_{21} extraction provides many high accuracy results. The best passive model, a series RL, provides a relatively poor fit over the 10 GHz bandwidth. To obtain a passive model, we require additional extractions. First, at low frequency, the series RL provides an excellent fit, Fig. 3. The impedance of this series RL is sub-

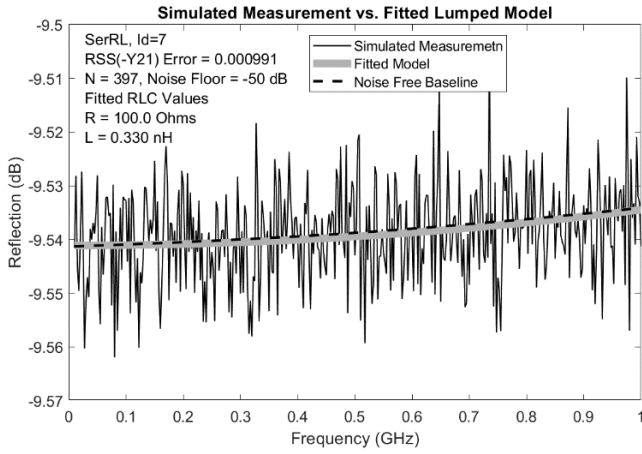


Fig. 3. Fitting only the low frequencies shows that a series RL provides the best passive model fit in the entire base library.

tracted from $1/y_{21}$, the result inverted, and a model extracted from the result. Essentially, we are extracting a model for y_{21} minus the low frequency model. After extracting a model, we re-attach the low frequency model. Subtracting and reattaching the low frequency admittance is also possible.

With the low frequency model removed, many suitable models are found for the remaining circuit. All include negative elements. However, these portions are often small resistors. When the 100 Ohm resistor from the low frequency model is reattached, it overwhelms the small non-passive resistor ($R_d = -0.04$ Ohms), yielding an overall passive model, Fig. 4. Fig. 5 and 6 show S-parameters from the resulting two-port model compared to the original, noise free baseline EM data.

VII. CONCLUSION

A least-squares fitting procedure for complex impedance or admittance data is specialized to rational polynomials with pure real coefficients yielding a perfectly causal transfer function. Synthesis equations provide extraction of RLC values for a library of 623 circuits. Perfectly passive circuits that result are easily identified and selected. The technique is validated with a known circuit and then illustrated with a model extraction for a two-port chip resistor.

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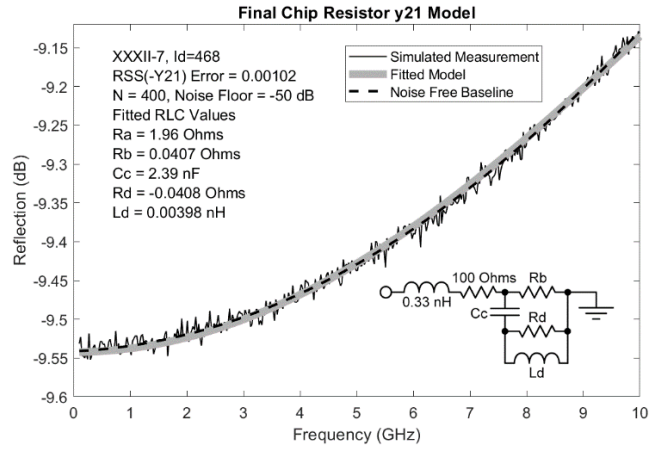


Fig. 4. The final fit for y_{21} was performed with the low frequency model subtracted, then added back in before forming the final y_{21} model.

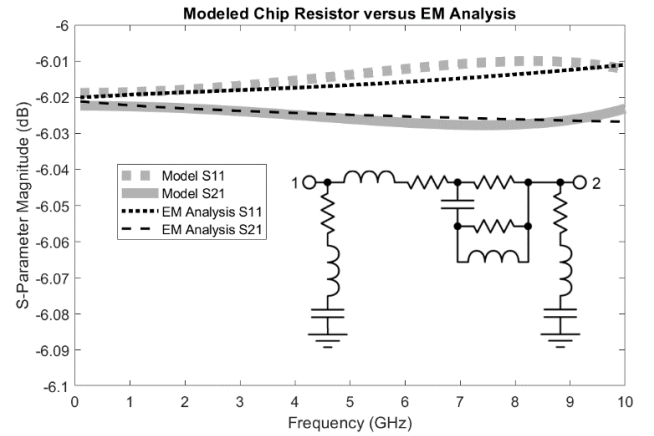


Fig. 5. The final two-port lumped model S-parameters magnitudes compared to the original EM analysis of the chip resistor. Note 0.01 dB per division.

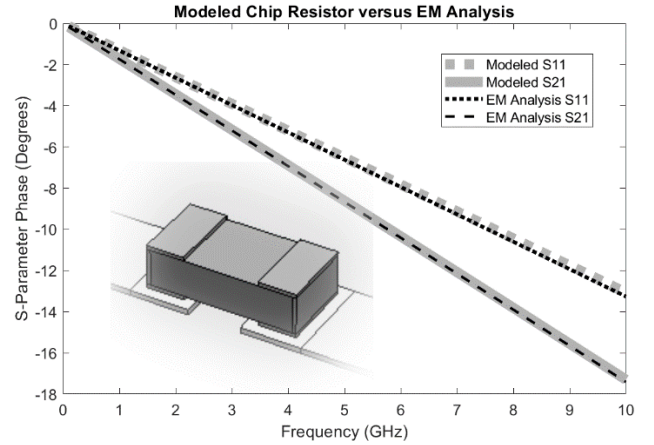


Fig. 6. Phase of the chip resistor model versus the original EM analysis.

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